

Introduction to Robotics

Personal Notes

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1 Intro to Robotics

1.1 Motivation and scope

Robotics brings computation into contact with the physical world. The opening examples range from household manipulation and humanoid locomotion to autonomous driving and drone delivery. They motivate a common goal: machines that perform useful tasks reliably outside a controlled demonstration.

The course develops the mathematical and algorithmic foundations of these systems, connecting machine learning, control, optimization, and computer vision. Deployment also raises questions about privacy, employment, safety, and whether a machine's behavior matches human intentions. These concerns accompany the technical problems rather than disappearing when a demonstration succeeds.

1.2 What counts as a robot?

The lecture places modern robotics in a longer history of imagined artificial beings and mechanical automata. Two landmarks are Karel Čapek's 1920 play *R.U.R.*, which introduced the word *robot*, and the development of the Unimate industrial manipulator by George Devol and Joseph Engelberger.

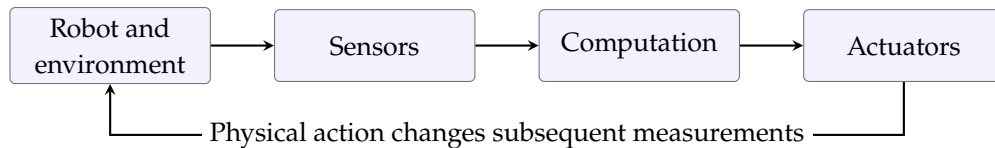
Definition: A working definition of a *robot* is a programmable physical agent whose behavior accomplishes tasks in its environment. Here, *embodiment* means that the agent has a physical form through which it interacts with the world.

Different definitions emphasize different properties: automatic execution of actions, reprogrammable manipulation, or the combination of sensing, computation, and physical action. These choices change which systems qualify. A dishwasher and an automatic door test how broad a definition should be; a surgical system controlled by a person tests whether autonomy is essential. A humanoid shape is not required, and programmability does not by itself imply independence from human supervision.

Note. There is no universally sharp boundary between robots and other machines. The working definition helps organize the subject; it is not a test that settles every borderline example.

1.3 The sensing–computation–action loop

A robotic system combines three functions. A *sensor* measures aspects of the robot or its surroundings. *Computation* converts measurements and task information into decisions. An *actuator* turns commands into physical effects, such as the thrust produced by a drone's motor-driven rotors. The resulting motion changes what the sensors observe next.



Original schematic of the lecture’s sensing–computation–action cycle.

The lecture uses a small quadrotor as a concrete example: its sensors include an inertial measurement unit and an optical-flow camera, its onboard computation processes information, and its rotors provide actuation. A *feedback* loop uses new measurements to adjust subsequent actions.

1.4 Three connected technical areas

Planning selects a course of action toward a goal; *control* determines how to execute and adjust that action. *Geometric motion planning* focuses on the spatial feasibility of motion, including obstacle avoidance. *Dynamics* describes how the robot’s motion evolves under applied inputs. The course moves from geometric planning to robot dynamics, feedback control, and planning subject to those dynamics.

State estimation infers quantities describing the robot from available measurements. *Localization* estimates where the robot is, while *mapping* estimates the surrounding environment. The course introduces Bayes filtering, Kalman filters, and particle filters as estimation methods. *Simultaneous localization and mapping (SLAM)* addresses localization and map construction together.

Robot learning uses data or experience to improve robotic behavior. The roadmap includes imitation learning, generative architectures, reinforcement learning, and world models. These are introduced here as topics for later lectures, not yet as algorithms with specified assumptions or guarantees.

1.5 Uncertainty as a common obstacle

Uncertainty is incomplete or imperfect knowledge relevant to a decision. The lecture connects it to all three technical areas:

Area	Sources of uncertainty
Planning and control	Robot dynamics, changes in the world, and the robot’s initial condition.
Localization and mapping	The robot’s state, environmental geometry, and the meaning of objects or observations.
Robot learning	Data quality, human intent, and the rewards supplied by interaction with the world.

Example: For a drone navigating around obstacles, a proposed route is useful only if the drone can execute it and estimate where it is. An imperfect motion model affects execution, while an inaccurate position estimate affects both control and obstacle avoidance. This illustrates why the three areas must work together.

1.6 Learning through implementation

The hardware platform is the Crazyflie 2.1 brushless quadrotor, a small, lightweight, open-source drone. The practical objective is autonomous navigation. Hardware work exposes the gap between a mathematical design and its behavior under real sensing and actuation. The lecture organizes this work in teams, with testing in designated netted spaces after the required lab training.

The assumed background is multivariable calculus, linear algebra, elementary probability, differential equations, and programming; the course uses Python. In the presented offering, problem sets account for 40% of the grade, the in-person midterm for 30%, and the final project for 30%. Assignments combine theory, code, and hardware. Collaboration is encouraged, but individual submissions remain the rule outside the team hardware portions.

The lecture's AI policy permits reviewing past assignments, explanations of technical concepts, Python syntax questions, and unrestricted use for hardware assignments and the final project. Its educational point is to develop skill with these tools while retaining an independent command of the fundamentals. Current administrative details belong to the course syllabus and Canvas.

There is no required textbook. Suggested references are *Principles of Robot Autonomy*, *Planning Algorithms*, and *Probabilistic Robotics*. ROB 549 is the graduate track and includes additional assignment problems. The technical sequence begins with geometric planning.

Source: Anirudha Majumdar, *Introduction to Robotics*, Princeton University, ROB 345 / 549, Lecture 1, "Introduction to Robotics," [lecture slides](#), PDF pages 1–57. [Official course materials](#); accessed September 19, 2026. These notes paraphrase the slides; the schematic and drone example are original explanatory additions.

2 Discrete planning (graph search)

2.1 The geometric planning model

The lecture isolates one part of autonomous navigation: finding a route through a known environment without collisions. This is the classical *piano mover's problem*, in which both the position and orientation of an extended object can matter. General formulations can be PSPACE-complete; practical methods therefore rely on useful representations and search strategies.

For now, assume that the environment map is given, the robot's configuration is known exactly, and every geometric path is feasible to execute. The last assumption sets aside dynamics: a collision-free route need not yet respect acceleration limits or the forces needed to follow it. Planning also appears outside navigation, for example when searching possible shapes of a molecule.

2.2 Configuration space

A *configuration* specifies the robot's geometric arrangement. Its *degrees of freedom (DoFs)* count the independent coordinates needed locally to describe that arrangement. The *configuration space* C is the set of all possible configurations. These coordinates do not include velocity in the geometric model.

System	DoFs	Configuration space
Point in d dimensions	d	$\mathbb{R}^d$
N independent points in d dimensions	Nd	$\mathbb{R}^{Nd}$
Pendulum with unrestricted angle	1	S^1
Planar quadrotor	3	$\mathbb{R}^2 \times S^1$

Here S^1 is the circle: angles differing by 2π describe the same orientation. For the planar quadrotor, $q = (x, y, \theta)$ combines two position coordinates with one orientation. Independent coordinates need not all have the geometry of an unbounded real line.

The *free configuration space* $C_{\text{free}} \subseteq C$ contains configurations without collision. Its complement, *obstacle configuration space*, is

$$C_{\text{obs}} = C \setminus C_{\text{free}}.$$

The planning problem is to find a continuous path

$$\gamma : [0, 1] \longrightarrow C_{\text{free}}, \quad \gamma(0) = q_A, \quad \gamma(1) = q_B.$$

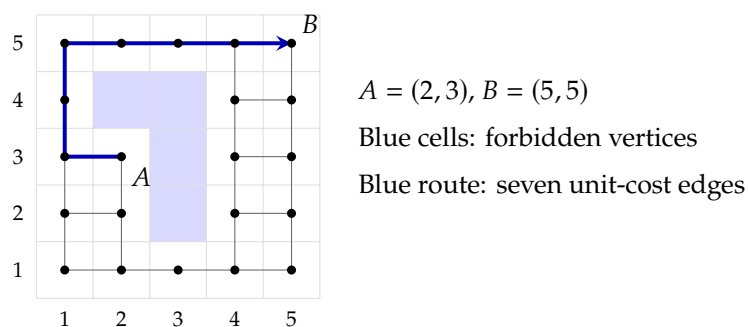
An obstacle in configuration space represents a forbidden arrangement of the whole robot. Checking only the position of a robot's center would ignore its extent and, when relevant, its orientation.

2.3 From a grid to a graph

Discretization replaces the continuous problem with candidate configurations and transitions. A *graph* $G = (V, E)$ records them as vertices V and edges E . Each grid cell supplies a vertex; adjacent cells are connected, and obstacle vertices and their incident edges are removed.

A *4-connected grid* permits horizontal and vertical moves. An *8-connected grid* also permits diagonal moves. The grid resolution and connectivity are modeling choices: a finer grid represents geometry more accurately but produces a larger search problem. A coarse grid can erase a narrow passage, so failure on the graph does not establish that no continuous path exists.

Note. The graph should represent collision-free motion, not just collision-free endpoints. Account for the robot's size when constructing obstacle cells, and check the transitions themselves; diagonal moves must not cut through obstacle corners. This is an implementation qualification to the grid abstraction.



Original rendering of the lecture's discrete example. Coordinates are (column, row).

2.4 Forward search and breadth-first search

Forward search explores reachable neighbors starting at A . Its *frontier* Q stores discovered vertices waiting to be processed. A vertex's *parent* records its discovering predecessor, enabling path recovery afterward.

The basic procedure is:

1. Insert A into Q and mark it discovered.
2. Remove a vertex x from Q . If x is a goal, recover its path and stop.
3. For each undiscovered neighbor y , set $\text{parent}(y) = x$, mark y discovered, and insert it into Q .
4. Repeat until a goal is removed or the frontier is empty. An empty frontier means that no goal is reachable in this graph.

Marking a vertex on insertion prevents repeated insertion through another edge or a cycle. To recover a path, follow parent pointers from the goal back to A and reverse the resulting sequence. A choice of frontier removal rule turns this procedure into different search algorithms.

Breadth-first search (BFS) uses a first-in, first-out queue: the oldest frontier vertex is processed next. Vertices are explored in layers of increasing number of edges from the start. For the illustrated graph and the lecture's neighbor order, the first few queue states are:

Vertex just processed	Queue afterward, next removal on the left
$A = (2, 3)$	$[(1, 3), (2, 2)]$
$(1, 3)$	$[(2, 2), (1, 4), (1, 2)]$
$(2, 2)$	$[(1, 4), (1, 2), (2, 1)]$

Proposition 2.1 [*BFS shortest paths*]. *On a finite graph, BFS finds a reachable goal using the fewest edges. This is also a minimum-cost path when every edge has the same positive cost.*

Proof. The queue processes every vertex at distance k before any vertex at distance $k + 1$. A vertex first discovered from layer k therefore has a path with $k + 1$ edges and cannot have an undiscovered shorter path. Applying this to the first goal processed proves the claim. $\square$

With adjacency lists and discovery checks, each vertex is inserted once and each edge is examined a bounded number of times, giving $O(|V| + |E|)$ time and $O(|V|)$ auxiliary storage. These bounds concern the discrete graph, whose size may itself be large.

2.5 Depth-first search

Depth-first search (DFS) instead uses a last-in, first-out stack: the most recently inserted vertex is processed next. It pursues a branch until it reaches the goal or runs out of new neighbors, then resumes another pending branch. Neighbor order matters because it determines which branch is tried first.

In the lecture's ordering, inserting (2, 2) and then (1, 3) makes (1, 3) the next vertex processed. DFS continues up the left side and across the top row, reaching B before exploring the remaining lower branch. Such behavior can be useful when a solution is deep and there are few promising branches.

On a finite graph, the discovered set still makes the search terminate, with $O(|V| + |E|)$ time and up to $O(|V|)$ stored information. DFS does not generally return a shortest path; finding a short route in this particular example is not an optimality guarantee.

2.6 Path costs and A* priorities

BFS minimizes the number of edges, not arbitrary travel costs, and may explore many vertices that do not lead toward the goal. Assign each edge (x, y) a nonnegative *edge cost* $\ell(x, y)$. The *path cost* for $p = (x_0, \dots, x_k)$ is

$$J(p) = \sum_{i=0}^{k-1} \ell(x_i, x_{i+1}).$$

The objective is now to minimize $J(p)$ among paths from A to B .

A search* orders the frontier using

$$F(x) = C(x) + H(x),$$

where the *cost-to-come* $C(x)$ is the cheapest cost to x found so far, and the *heuristic* $H(x)$ estimates the remaining cost to the goal. Thus $C(x)$ is the cost of a known path, not necessarily the optimal cost to x yet. The frontier is a *priority queue*, which removes a vertex with the smallest F value.

An *admissible heuristic* never exceeds the true minimum remaining cost $H^*(x)$: $0 \leq H(x) \leq H^*(x)$, with $H(B) = 0$. Ignoring obstacles often produces a useful lower bound. Euclidean distance is valid when each edge's cost is at least its Euclidean displacement. Setting $H \equiv 0$ gives *Dijkstra's algorithm*, which prioritizes cost-to-come alone.

For the lecture's 4-connected grid with unit edge costs, the *Manhattan distance*

$$H(x) = |x_1 - 5| + |x_2 - 5|$$

counts the horizontal and vertical moves needed without obstacles. Obstacles can require detours but cannot reduce that count. Initially $C(A) = 0$, $H(A) = 5$, and $F(A) = 5$. The first updates are:

Newly reached vertex	C	H	F
(1, 3), from A	1	6	7
(2, 2), from A	1	6	7
(1, 4), from (1, 3)	2	5	7
(1, 2), from (1, 3)	2	7	9
(1, 5), from (1, 4)	3	4	7

The first two candidates tie in both F and H ; either may be chosen. Later, breaking an F tie by smaller H favors $(1, 4)$ over $(2, 2)$ and directs search along the upper route. The illustrated solution costs 7, exceeding the initial lower bound 5 because of the obstacle.

Heuristics must match the motion and cost model. Manhattan distance can overestimate when diagonal moves are allowed cheaply. The lecture also points to learned search heuristics; a learned estimate does not automatically satisfy the lower-bound condition required for an optimality guarantee.

2.7 A* updates and implementation

The lecture distinguishes undiscovered vertices, *open vertices* (alive, stored in the frontier), and *closed vertices* (dead, already expanded). Initialize $C(A) = 0$ and $C(x) = \infty$ otherwise. When expanding x , compute the candidate cost $C(x) + \ell(x, y)$ for each neighbor y . If it improves $C(y)$, update $C(y)$, its parent, and its priority. This operation is *relaxation*. Unlike BFS discovery marking, first encountering a vertex does not settle its cheapest path.

Note. **Qualification to the slides' permanent closed set.** A *consistent heuristic* satisfies $H(x) \leq \ell(x, y) + H(y)$ on every edge, with $H(B) = 0$. Consistency ensures that closing a vertex does not discard a later cheaper route. Admissibility alone does not justify this rule: with an inconsistent heuristic, improved paths may require reopening closed vertices. Manhattan distance on this unit-cost, 4-connected grid is consistent.

The implementation below follows the permanent-closure version. It assumes a finite graph, nonnegative costs, and a consistent nonnegative heuristic that is zero at the goal. The function `neighbors(x)` yields pairs (vertex, edge_cost). Improved priorities are pushed as new heap entries; obsolete entries are ignored when removed. A serial counter resolves remaining ties without comparing vertex identifiers.

Python implementation

```
1 from heapq import heappop, heappush
2 from itertools import count

3 def astar(start, goal, neighbors, heuristic):
4     serial = count()
5     h0 = heuristic(start)
6     queue = [(h0, h0, next(serial), 0, start)]
7     cost, parent, closed = {start: 0}, {}, set()
8     while queue:
9         _, _, _, g, x = heappop(queue)
10        if x in closed or g != cost[x]:
11            continue
12        if x == goal:
13            path = [x]
14            while x != start:
15                x = parent[x]
16                path.append(x)
17            return path[::-1], g
18        closed.add(x)
19        for y, edge_cost in neighbors(x):
20            if y in closed:
21                continue
22            candidate = g + edge_cost
23            if candidate < cost.get(y, float("inf")):
24                cost[y], parent[y] = candidate, x
25                h = heuristic(y)
26                heappush(queue, (candidate + h, h,
27                                next(serial), candidate, y))
28    return None
```

Stop when the goal is removed as the best valid frontier entry, not when it is first discovered. If the frontier becomes empty, no path exists in the graph. Even a minimum-cost graph path is optimal only for the chosen discretization, edge costs, and geometric assumptions.

Source: Anirudha Majumdar, *Introduction to Robotics*, Princeton University, ROB 345 / 549, Lecture 2, “Discrete Motion Planning,” [lecture slides](#), PDF pages 1–79, listed as “Discrete planning (graph search)” on the [official course page](#); accessed September 20, 2026. The diagram, proof, and Python implementation are original explanatory reconstructions. The consistency qualification is supported by David L. Poole and Alan K. Mackworth, *Artificial Intelligence: Foundations of Computational Agents*, 2nd ed., [Section 3.7.2](#).